KAIZEN MATHS

Lesson starter

Integration of Algebraic Fractions

Integrating Partial Fractions

Name _____

S1

Decompose into partial fractions, then integrate.

Find $\int \frac{3x+23}{(x+1)(x+5)}$ dx.

S2

Use the repeated-factor partial fraction form, then integrate.

Find $\int \frac{4x+20}{(x+6)^2}$ dx.

S3

Divide first, decompose the proper fraction, then integrate.

Find $\int \frac{4x^2+11x+9}{(x+1)(x+2)}$ dx.

S4

Decompose into partial fractions, then integrate.

Find $\int \frac{3x+18}{(x+4)(x+7)}$ dx.

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ALG-31E3-L3 · S

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KAIZEN MATHS

Exit ticket

Integration of Algebraic Fractions

Integrating Partial Fractions

Name _____

E1

Use the repeated-factor partial fraction form, then integrate.

Find $\int \frac{3x+6}{(x+3)^2}\,dx$.

E2

Divide first, decompose the proper fraction, then integrate.

Find $\int \frac{x^2+9x+12}{(x+1)(x+3)}\,dx$.

E3

Decompose into partial fractions, then integrate.

Find $\int \frac{-7x-22}{(x+4)(x+1)}\,dx$.

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Teacher answers & working

Integration of Algebraic Fractions · Integrating Partial Fractions

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S1

Decompose into partial fractions, then integrate.

Find $\int \frac{3x+23}{(x+1)(x+5)}\,dx$.

Answer:

$5\ln|x+1| - 2\ln|x+5| + C$

Working:

Decompose the fraction first.

$\frac{3x+23}{(x+1)(x+5)} = \frac{A}{x+1} + \frac{B}{x+5}$.

$3x + 23 = A(x + 5) + B(x + 1)$.

Put $x = -1$: $20 = A(4) + B(0)$.

$A = 5 = 5$.

Put $x = -5$: $8 = A(0) + B(-4)$.

$B = -2 = -2$.

So $\frac{3x+23}{(x+1)(x+5)} = \frac{5}{x+1} + \frac{-2}{x+5}$.

$\int \frac{5}{x+1}\,dx = 5\ln|x+1|$.

$\int \frac{-2}{x+5}\,dx = -2\ln|x+5|$.

Combine: $5\ln|x+1| - 2\ln|x+5| + C$.

Teacher answers & working

Integration of Algebraic Fractions · Integrating Partial Fractions

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S2

Use the repeated-factor partial fraction form, then integrate.

Find $\int \frac{4x+20}{(x+6)^2}\,dx$.

Answer:

$4\ln|x+6| + 4\frac{1}{x+6} + C$

Working:

Use the repeated-factor form.

$\frac{4x+20}{(x+6)^2} = \frac{A}{x+6} + \frac{B}{(x+6)^2}$.

$4x + 20 = A(x+6) + B$.

Expand: $4x + 20 = Ax + 6A + B$.

Compare x-coefficients: $A = 4$.

Compare constants: $20 = 6(4) + B$.

$B = 20 - (24) = -4$.

$\int \frac{4}{x+6}\,dx = 4\ln|x+6|$.

Rewrite: $\frac{-4}{(x+6)^2} = -4(x+6)^{-2}$.

New power: $-2 + 1 = -1$.

Divide by -1: $\frac{-4}{-1}(x+6)^{-1}$.

$\int \frac{-4}{(x+6)^2}\,dx = 4\frac{1}{x+6}$.

Final answer: $4\ln|x+6| + 4\frac{1}{x+6} + C$.

Teacher answers & working

Integration of Algebraic Fractions · Integrating Partial Fractions

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S3

Divide first, decompose the proper fraction, then integrate.

Find $\int \frac{4x^2+11x+9}{(x+1)(x+2)}\,dx$.

Answer:

$4x + 2\ln|x + 1| - 3\ln|x + 2| + C$

Working:

The fraction is improper, so divide or rewrite it first.

$\frac{4x^2+11x+9}{(x+1)(x+2)} = 4 + \frac{-x+1}{(x+1)(x+2)}$.

Let $\frac{-x+1}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2}$.

$-x + 1 = A(x + 2) + B(x + 1)$.

Put $x = -1$: $2 = A(1)$.

$A = 2$.

Put $x = -2$: $3 = B(-1)$.

$B = -3$.

$\frac{-x+1}{(x+1)(x+2)} = \frac{2}{x+1} + \frac{-3}{x+2}$.

So the integrand is $4 + \frac{2}{x+1} + \frac{-3}{x+2}$.

$\int 4\,dx = 4x$.

$\int \frac{2}{x+1}\,dx = 2\ln|x + 1|$.

$\int \frac{-3}{x+2}\,dx = -3\ln|x + 2|$.

Integrate term by term: $4x + 2\ln|x + 1| - 3\ln|x + 2| + C$.

Teacher answers & working

Integration of Algebraic Fractions · Integrating Partial Fractions

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S4

Decompose into partial fractions, then integrate.

Find $\int \frac{3x+18}{(x+4)(x+7)}\,dx$.

Answer:

$2\ln|x+4| + \ln|x+7| + C$

Working:

Decompose the fraction first.

$\frac{3x+18}{(x+4)(x+7)} = \frac{A}{x+4} + \frac{B}{x+7}$.

$3x + 18 = A(x + 7) + B(x + 4)$.

Put $x = -4$: $6 = A(3) + B(0)$.

$A = 2 = 2$.

Put $x = -7$: $-3 = A(0) + B(-3)$.

$B = 1 = 1$.

So $\frac{3x+18}{(x+4)(x+7)} = \frac{2}{x+4} + \frac{1}{x+7}$.

$\int \frac{2}{x+4}\,dx = 2\ln|x+4|$.

$\int \frac{1}{x+7}\,dx = 1\ln|x+7|$.

Combine: $2\ln|x+4| + \ln|x+7| + C$.

Teacher answers & working

Integration of Algebraic Fractions · Integrating Partial Fractions

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E1

Use the repeated-factor partial fraction form, then integrate.

Find $\int \frac{3x+6}{(x+3)^2}\,dx$.

Answer:

$3\ln|x+3| + 3\frac{1}{x+3} + C$

Working:

Use the repeated-factor form.

$\frac{3x+6}{(x+3)^2} = \frac{A}{x+3} + \frac{B}{(x+3)^2}$.

$3x + 6 = A(x + 3) + B$.

Expand: 3x + 6 = Ax + 3A + B.

Compare x-coefficients: A = 3.

Compare constants: 6 = 3(3) + B.

$B = 6 - (9) = -3$.

$\int \frac{3}{x+3}\,dx = 3\ln|x+3|$.

Rewrite: $\frac{-3}{(x+3)^2} = -3(x+3)^{-2}$.

New power: −2 + 1 = −1.

Divide by −1: $\frac{-3}{-1}(x+3)^{-1}$.

$\int \frac{-3}{(x+3)^2}\,dx = 3\frac{1}{x+3}$.

Final answer: $3\ln|x+3| + 3\frac{1}{x+3} + C$.

Teacher answers & working

Integration of Algebraic Fractions · Integrating Partial Fractions

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E2

Divide first, decompose the proper fraction, then integrate.

Find $\int \frac{x^2+9x+12}{(x+1)(x+3)}\,dx$.

Answer:

$1x + 2\ln|x + 1| + 3\ln|x + 3| + C$

Working:

The fraction is improper, so divide or rewrite it first.

$\frac{x^2+9x+12}{(x+1)(x+3)} = 1 + \frac{5x+9}{(x+1)(x+3)}$.

Let $\frac{5x+9}{(x+1)(x+3)} = \frac{A}{x+1} + \frac{B}{x+3}$.

$5x + 9 = A(x + 3) + B(x + 1)$.

Put $x = -1$: $4 = A(2)$.

$A = 2$.

Put $x = -3$: $-6 = B(-2)$.

$B = 3$.

$\frac{5x+9}{(x+1)(x+3)} = \frac{2}{x+1} + \frac{3}{x+3}$.

So the integrand is $1 + \frac{2}{x+1} + \frac{3}{x+3}$.

$\int 1\,dx = x$.

$\int \frac{2}{x+1}\,dx = 2\ln|x + 1|$.

$\int \frac{3}{x+3}\,dx = 3\ln|x + 3|$.

Integrate term by term: $x + 2\ln|x + 1| + 3\ln|x + 3| + C$.

Teacher answers & working

Integration of Algebraic Fractions · Integrating Partial Fractions

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E3

Decompose into partial fractions, then integrate.

Find $\int \frac{-7x-22}{(x+4)(x+1)}\,dx$.

Answer:

$-2\ln|x+4| - 5\ln|x+1| + C$

Working:

Decompose the fraction first.

$\frac{-7x-22}{(x+4)(x+1)} = \frac{A}{x+4} + \frac{B}{x+1}$.

$-7x - 22 = A(x+1) + B(x+4)$.

Put $x = -4$: $6 = A(-3) + B(0)$.

$A = -2 = -2$.

Put $x = -1$: $-15 = A(0) + B(3)$.

$B = -5 = -5$.

So $\frac{-7x-22}{(x+4)(x+1)} = \frac{-2}{x+4} + \frac{-5}{x+1}$.

$\int \frac{-2}{x+4}\,dx = -2\ln|x+4|$.

$\int \frac{-5}{x+1}\,dx = -5\ln|x+1|$.

Combine: $-2\ln|x+4| - 5\ln|x+1| + C$.